

Table 7 Logical Equivalences Involving Conditional Statements

- ① $p \rightarrow q \equiv \neg p \vee q$ ② $p \rightarrow q \equiv \neg q \rightarrow \neg p$ ③ $p \vee q \equiv \neg p \rightarrow q$
④ $p \wedge q \equiv \neg(p \rightarrow \neg q)$ ⑤ $\neg(p \rightarrow q) \equiv p \wedge \neg q$ ⑥ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
⑦ $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ ⑧ $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
⑨ $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Table 8 Logical Equivalences Involving Biconditional Statements

- ① $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ ② $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ ③ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ ④ $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

We will verify the equivalences in Tables 6-8 in the exercises.

1.3.3. Using De Morgan's Laws

Example 5: Use De Morgan's laws to express the negations of "Miguel has a cellphone and he has a laptop computer" and "Heather will go to the concert or Steve will go to the concert."

Soln: p : Miguel has a cellphone q : Miguel has a laptop computer.

The statement given can be expressed as $p \wedge q$.
By De Morgan's law, $\neg(p \wedge q) \equiv \neg p \vee \neg q$. This can be expressed in English as "Miguel does not have a cellphone or he does not have a laptop computer."

r : Heather will go to the concert s : Steve will go to the concert.

The statement given can be expressed as $(r \vee s)$.
By De Morgan's law, $\neg(r \vee s) \equiv \neg r \wedge \neg s$. This can be expressed in English as "Heather will not go to the concert and Steve will not go to the concert."

1.3.4 Constructing New Logical Equivalences

The logical equivalences in Table 6, as well as any others that have been established (such as those shown in Tables 7 and 8), can be used to construct additional logical equivalences. The reason for this is that a proposition in a compound proposition can be replaced by a compound proposition that is logically equivalent to it without changing the truth.